



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO THIRUMALAI THIRUPARAI AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

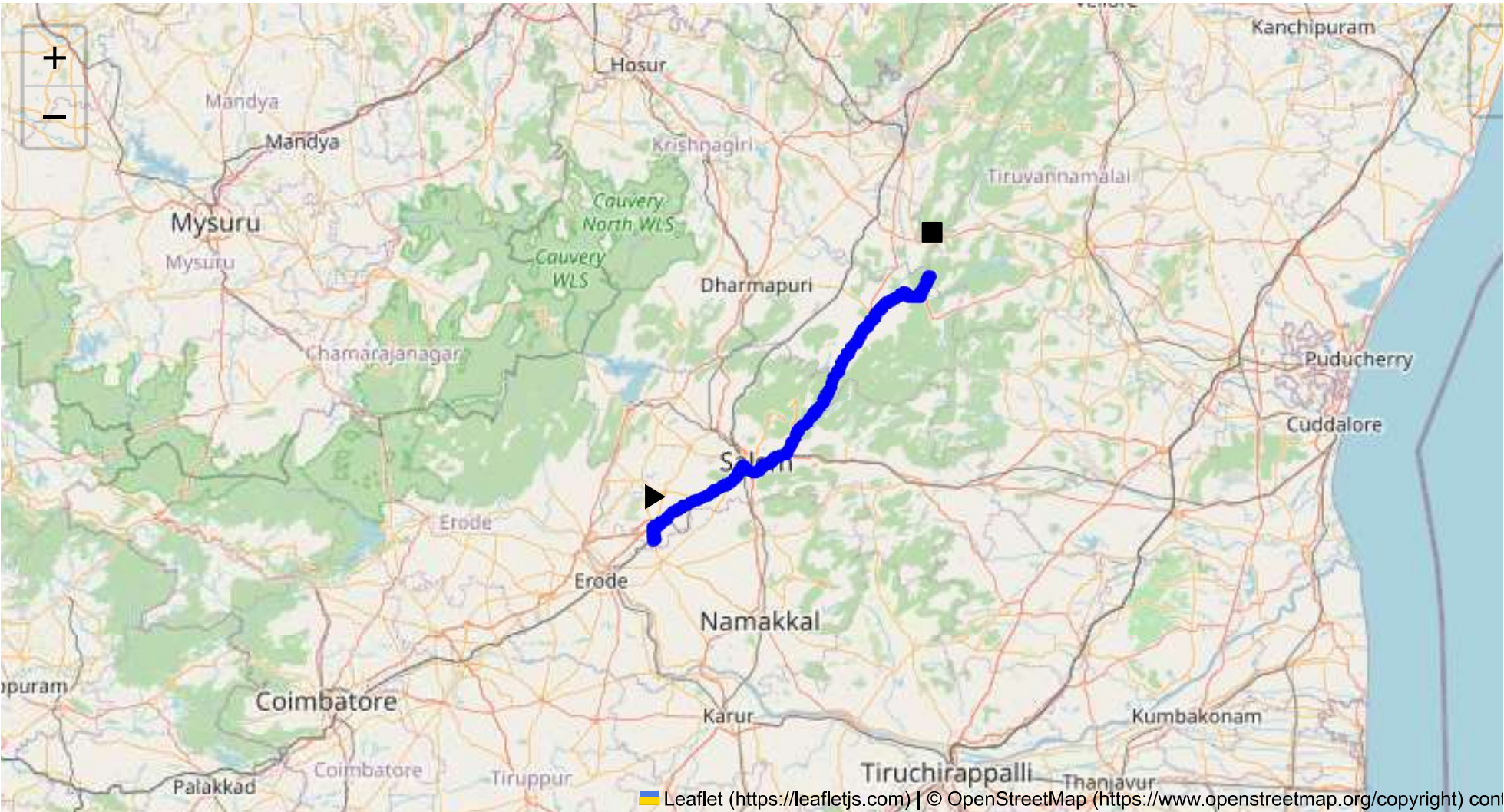
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 129.91 km
Estimated Duration: 3.1 hours
Adjusted Duration (Heavy Vehicle): 3.9 hours
Start: (11.4381, 77.8734)
End: (12.158, 78.638282)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 5J5Q+578, Tambal spans approximately 129.91 kilometers, typically traversing through major roads such as NH44 and SH20. The route takes the driver from a mix of urban and semi-urban environments into more rural settings.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences tropical weather with three main seasons: summer, monsoon, and winter. Key weather-related hazards along the route include:

- **Heavy Rainfall:** During the monsoon (June to September), roads can become slippery and visibility may reduce, increasing the risk of accidents.
- **Heat Waves:** In peak summer months (March to May), extreme heat can impact vehicle performance and driver endurance.
- **Fog and Mist:** During late night and early mornings in winter (November to January), visibility may be compromised.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (7:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) are typical peak times due to work commutes and school traffic.
- **Congestion-Prone Areas:** Sangagiri town center, entry, and exit points along NH44, and areas near Salem city are particularly prone to congestion.

4. Assessment of Road Quality and Infrastructure

- **NH44:** Generally in good condition, well-marked, and with adequate signage. However, road work or patches may occur periodically.
- **SH20:** Road quality may vary, with some stretches being narrower and more prone to potholes, especially post-monsoon.

5. Suggestions for Alternative Routes

In emergencies, consider diverting traffic through Salem's city roads or using local roads through Erode district if conditions allow, keeping in mind traffic heavy routes may further delay travel.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permit Requirements:** Ensure all necessary permits for hazardous material shipments are in place.
- **Time Restrictions:** Some urban regions may have restrictions on the timing of heavy vehicle movement.
- **Safety Protocols:** Adherence to safety protocols including signage on vehicles, use of protective gear, and emergency response plans is critical.

7. Overview of Historical Incidents

Historically, accidents involving heavy vehicles on NH44 have been due to vehicular collisions often attributed to overtaking and speeding. Awareness of local accident black spots is advised for preventative

measures.

8. Environmental Considerations and Sensitive Areas

- **Reserved Forests and Wildlife Zones:** Certain parts might pass near protected areas. Ensuring no leakage or spillage is vital.
- **Water Bodies:** Mismanagement or accidents near lakes and rivers could lead to severe environmental damage.

9. Analysis of Communication Coverage

Most of the route is expected to have good mobile coverage due to proximity to NH44. However, remote rural stretches and forest areas may experience dropped signals.

10. Estimated Emergency Response Times

- **Urban Areas:** Approximately 20-30 minutes due to better road access and proximity to emergency services.
- **Rural Areas:** 40-60 minutes, depending on the specific location and accessibility.

11. Overall Summary of Risk Assessment

This route presents moderate overall risk for heavy vehicles transporting hazardous materials. Traffic congestion, weather conditions, and road quality are the primary concerns. Preparedness with legal compliance, travel during off-peak hours, and careful attention in high-risk areas will help mitigate potential risks.

The analysis underscores the importance of monitoring live traffic updates and weather conditions to adjust travel plans accordingly. Improved driver training on emergency response and adherence to safety standards are essential to maintain safety on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Roundabout	High	12.11402, 78.56990	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.53501, 77.95707	30 KM/Hr
12	Turn	Medium	11.53601, 77.95856	30 KM/Hr
13	Turn	Medium	11.62403, 78.10893	30 KM/Hr
14	Turn	High	11.64172, 78.11817	15 KM/Hr
15	Turn	Medium	11.64224, 78.11928	30 KM/Hr
16	Turn	High	11.64195, 78.12129	15 KM/Hr
17	Turn	High	11.64120, 78.12168	15 KM/Hr
18	Turn	Medium	11.63948, 78.12081	30 KM/Hr
19	Turn	Medium	11.62861, 78.16526	30 KM/Hr
20	Turn	Medium	11.65866, 78.20494	30 KM/Hr
21	Turn	Medium	11.66388, 78.20672	30 KM/Hr
22	Turn	Medium	11.67129, 78.22978	30 KM/Hr
23	Turn	Medium	11.67116, 78.23102	30 KM/Hr
24	Turn	High	11.67145, 78.23141	15 KM/Hr
25	Turn	High	11.67198, 78.23148	15 KM/Hr
26	Turn	Medium	11.67440, 78.23914	30 KM/Hr
27	Turn	Medium	11.80440, 78.33325	30 KM/Hr
28	Turn	Medium	11.81235, 78.34084	30 KM/Hr
29	Turn	Medium	11.83482, 78.35294	30 KM/Hr
30	Turn	Medium	11.84738, 78.36062	30 KM/Hr
31	Turn	Medium	11.99554, 78.44553	30 KM/Hr
32	Turn	Medium	12.01577, 78.45673	30 KM/Hr
33	Turn	Medium	12.04335, 78.48368	30 KM/Hr
34	Blind Spot	Blind Spot	12.04669, 78.48390	10 KM/Hr
35	Turn	Medium	12.04654, 78.48540	30 KM/Hr
36	Turn	Medium	12.04700, 78.48599	30 KM/Hr
37	Turn	Medium	12.07921, 78.50963	30 KM/Hr
38	Turn	Medium	12.10411, 78.55184	30 KM/Hr
39	Turn	Medium	12.10219, 78.58961	30 KM/Hr
40	Turn	Medium	12.10223, 78.59080	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
41	Turn	Medium	12.10440, 78.60609	30 KM/Hr
42	Turn	Medium	12.10492, 78.60709	30 KM/Hr
43	Turn	High	12.10426, 78.61172	15 KM/Hr
44	Turn	Medium	12.14980, 78.63214	30 KM/Hr
45	Turn	Medium	12.15610, 78.63606	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
21	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
22	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
23	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
24	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
25	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
26	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
27	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
28	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
29	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
31	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
33	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
34	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
20	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
30	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
32	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
35	school	Indian Matriculation Higher secondary school	12.006299, 78.4531204	30 km/h	Medium
36	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
37	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
38	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
39	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
40	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
41	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.63303, 78.12713



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.11402, 78.56990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64224, 78.11928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64195, 78.12129



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62861, 78.16526



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65866, 78.20494



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66388, 78.20672



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67129, 78.22978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67116, 78.23102



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67145, 78.23141



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67198, 78.23148



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67440, 78.23914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80440, 78.33325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81235, 78.34084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83482, 78.35294



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84738, 78.36062



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99554, 78.44553



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01577, 78.45673



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04335, 78.48368



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.04669, 78.48390



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04654, 78.48540



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04700, 78.48599



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07921, 78.50963



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.10411, 78.55184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.10219, 78.58961



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.10223, 78.59080



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.10492, 78.60709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.10426, 78.61172